

Prince Gulia

Backend Developer

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Profile

Backend Developer and BCA student specializing in high-performance REST APIs, real-time architectures, and AI-integrated systems. Proven experience building asynchronous background queues (BullMQ/Redis), JWT authentication, and secure, document-grounded RAG pipelines using pgvector. Dedicated to optimizing backend efficiency, resource utilization, and system performance.

Education

Bachelor of Computer Applications (BCA)

2024 – Present | New Delhi, India

Guru Gobind Singh Indraprastha University (IINTM Janakpuri)

- GPA: 9.5 / 10.0
- Coursework: Data Structures & Algorithm, DBMS, Web Development, Operating System

Technical Skills

- **Languages** — JavaScript, Python, C++, SQL
- **Backend** — Node.js, Express.js, REST APIs, WebSockets, Socket.io, Redis
- **Databases** — PostgreSQL, pgvector
- **Auth & Security** — JWT, Bcrypt
- **AI/ML** — GenAI, RAG Pipelines, Vector Embeddings
- **Tools** — Git, GitHub, Streamlit, Linux

Projects

DocuMind - AI Document Assistant [🔗](#)

Node.js/Express, PostgreSQL, pgvector, BullMQ, Gemini API, RAG

- Architected a RAG pipeline utilizing the Gemini API for vector embeddings and pgvector for high-performance semantic search.
- Engineered an asynchronous background queue using BullMQ and ioredis for PDF processing, maintaining high API responsiveness.
- Optimized search accuracy with a 500-word text chunking algorithm, utilizing boundary overlaps to prevent context loss.
- Designed strict AI query guardrails to guarantee document-grounded responses and eliminate out-of-scope hallucinations.

Real Time Chat App [🔗](#)

Node.js/Express, Socket.io, PostgreSQL, JWT, REST APIs

- Built a real-time chat backend with Socket.io, handling persistent WebSocket connections authenticated via JWT.
- Optimized message payloads and connection events to ensure sub-second message delivery within strict free-tier memory constraints (512MB RAM).
- Engineered a PostgreSQL schema for message history, utilizing efficient indexing to maintain high API responsiveness under limited server resources.
- Implemented secure room-based broadcasting, isolating chat events and ensuring accurate relational room-user mapping.

Study Tracker - Habit Tracker App [🔗](#)

Node.js/Express, Chart.js, JWT, REST APIs

- Developed JWT-authenticated study tracking platform supporting task categorization and progress monitoring.
- Engineered REST APIs for CRUD operations with secure user-specific data access.
- Generated productivity analytics using Chart.js visualizations and aggregated study statistics.
- Designed modular backend architecture for future feature scalability.